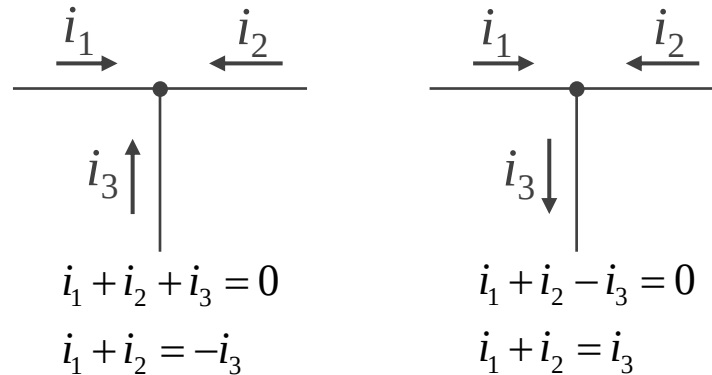


Topic 2 recap

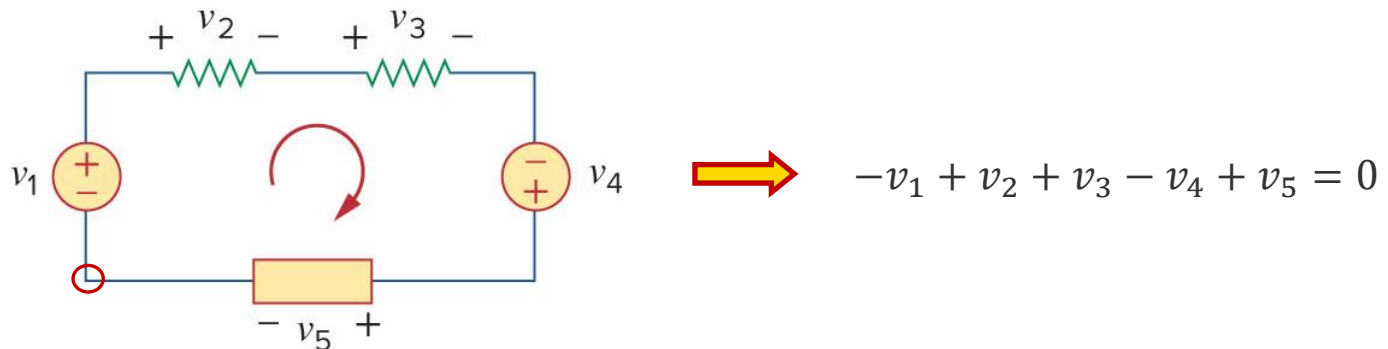
- Kirchhoff's Current Law (KCL):

- The sum of all current entering and leaving a node is zero $\sum_{n=1}^N i_n = 0$.



- Kirchhoff's Voltage Law (KVL):

- The sum of all voltages in a loop is zero $\sum_{m=1}^M v_m = 0$.



Topic 2 recap

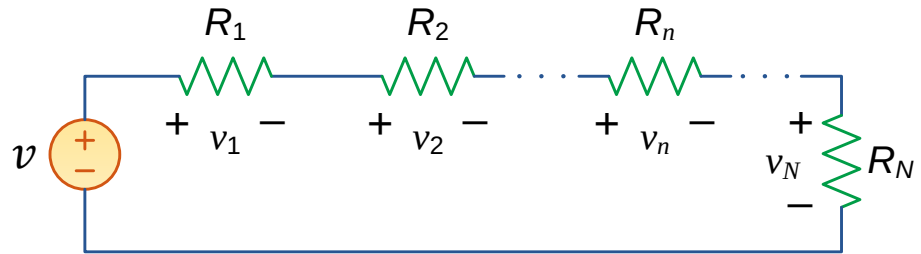
Circuits can be simplified by finding the equivalent resistance from a given terminal.

- Resistors in series: $R_{eq} = R_1 + R_2 + \dots + R_N$

- Resistors in parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$

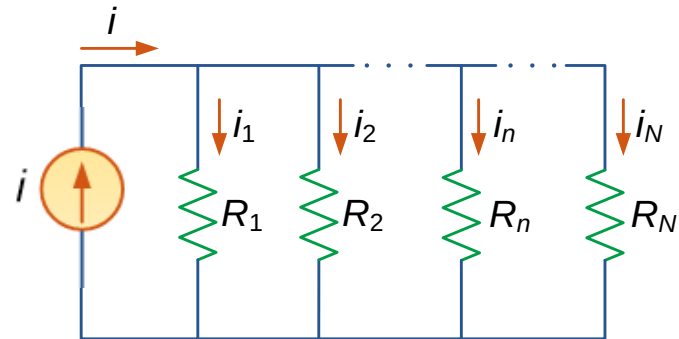
- Voltage division:

$$v_n = v \frac{R_n}{R_1 + R_2 + \dots + R_N} = v \frac{R_n}{R_{eq}}$$



- Current division:

$$i_n = i \frac{R_1 \parallel R_2 \parallel \dots \parallel R_N}{R_n} = i \frac{R_{eq}}{R_n}$$



Topic 2 recap – Nodal analysis

1. Choose **ground** as **reference node** ($v_0 = 0$ V).
2. Assign voltages v_1 and v_2 to nodes 1 and 2.
3. Apply KCL to nodes 1 and 2.

node 1: $I_1 = I_2 + i_1 + i_2$

node 2: $I_2 + i_2 = i_3$

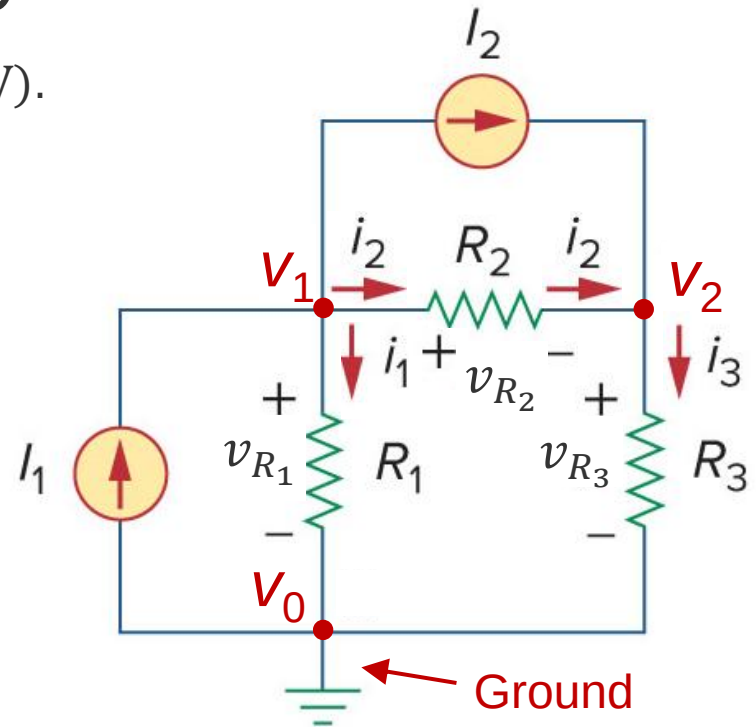
- For resistors, use Ohm's law to express the branch currents in terms of node voltages. Keep in mind the passive sign convention.

$$i_n = \frac{v_{R_n}}{R_n} = \frac{v_{\text{higher}} - v_{\text{lower}}}{R_n}$$

$$i_1 = \frac{v_{R_1}}{R_1} = \frac{v_1 - 0}{R_1}; \quad i_2 = \frac{v_{R_2}}{R_2} = \frac{v_1 - v_2}{R_2}; \quad i_3 = \frac{v_{R_3}}{R_3} = \frac{v_2 - 0}{R_3}$$

- Substitute back i_1 , i_2 , and i_3 into the node equations

4. Solve simultaneous equations (for v_1 and v_2).



$$\begin{cases} I_1 = I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2} \\ I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2}{R_3} \end{cases}$$

Topic 2 recap – Nodal analysis

- Nodal analysis with voltage source:

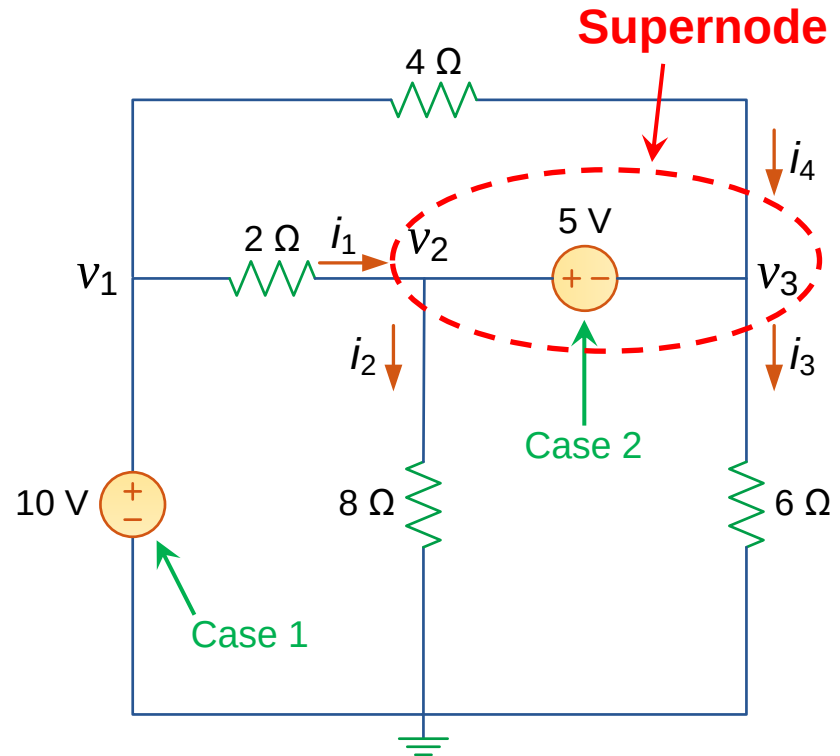
- Case 1: Voltage source is connected to the reference node.

$$v_k = v_{\text{source}}$$

- Case 2: Voltage source is between two non-reference nodes.

- Form a supernode by enclosing voltage source and any parallel element with it.
- Write the constraint equation relating the node voltages inside the supernode.

$$v_a - v_b = v_{\text{source}}$$



Topic 2 recap – Mesh analysis

1. Assign currents i_1 and i_2 to meshes 1 and 2.
2. Apply KVL to meshes 1 and 2.

$$\text{mesh 1: } -V_1 + v_{R_1} + v_{R_3} = 0$$

$$\text{mesh 2: } -v_{R_3} + v_{R_2} + V_2 = 0$$

- For resistors, use Ohm's law to express the voltages in terms of mesh currents.

$$v_{R_1} = R_1 I_1 = R_1 i_1$$

$$v_{R_2} = R_2 I_2 = R_2 i_2$$

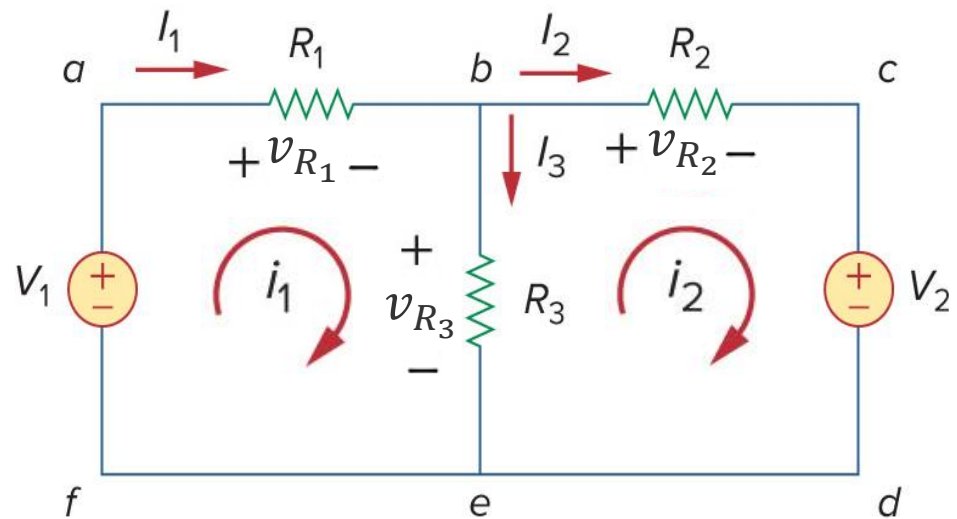
$$v_{R_3} = R_3 I_3 = R_3(i_1 - i_2)$$

- Substitute back v_{R_1} , v_{R_2} , and v_{R_3} into the mesh equations.



$$\begin{cases} -V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0 \\ -R_3(i_1 - i_2) + R_2 i_2 + V_2 = 0 \end{cases}$$

3. Solve simultaneous equations (for i_1 and i_2).



Topic 2 recap – Mesh analysis

- Mesh analysis with current source:

- Case 1: Current source is only in one mesh.

$$i_k = i_{\text{source}}$$

- Case 2: Current source is shared between two meshes.

- Form a supermesh by excluding the current source and any series element with it.
- Write the constraint equation relating the mesh currents inside the supermesh.

$$i_a - i_b = i_{\text{source}}$$

